

Zomato Data Analysis Using Python.

Import all the necessary python libraries.

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

load the file and printing also

```
In [2]: df = pd.read_csv("zomato data .csv")
print(df)
```

	name	online_order	book_table	rate	votes	\
0	Jalsa	Yes	Yes	4.1/5	775	
1	Spice Elephant	Yes	No	4.1/5	787	
2	San Churro Cafe	Yes	No	3.8/5	918	
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	
4	Grand Village	No	No	3.8/5	166	
..	...	...	...	...	...	
143	Melting Melodies	No	No	3.3/5	0	
144	New Indraprasta	No	No	3.3/5	0	
145	Anna Kuteera	Yes	No	4.0/5	771	
146	Darbar	No	No	3.0/5	98	
147	Vijayalakshmi	Yes	No	3.9/5	47	

	approx_cost(for two people)	listed_in(type)
0	800	Buffet
1	800	Buffet
2	800	Buffet
3	300	Buffet
4	600	Buffet
..	...	...
143	100	Dining
144	150	Dining
145	450	Dining
146	800	Dining
147	200	Dining

[148 rows x 7 columns]

```
In [3]: df
```

Out[3]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1/5	775	800	Buffet
1	Spice Elephant	Yes	No	4.1/5	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8/5	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300	Buffet
4	Grand Village	No	No	3.8/5	166	600	Buffet
...	...	...	...	...	...	...	...
143	Melting Melodies	No	No	3.3/5	0	100	Dining
144	New Indraprasta	No	No	3.3/5	0	150	Dining
145	Anna Kuteera	Yes	No	4.0/5	771	450	Dining
146	Darbar	No	No	3.0/5	98	800	Dining
147	Vijayalakshmi	Yes	No	3.9/5	47	200	Dining

148 rows × 7 columns

Convert the data type of the "rate" column to float and remove denominator.

```
In [4]: def rate(value):
        value=str(value).split('/')
        value=value[0];
        return float(value)

df['rate']=df['rate'].apply(rate)
print(df.head())
```

	name	online_order	book_table	rate	votes	\
0	Jalsa	Yes	Yes	4.1	775	
1	Spice Elephant	Yes	No	4.1	787	
2	San Churro Cafe	Yes	No	3.8	918	
3	Addhuri Udupi Bhojana	No	No	3.7	88	
4	Grand Village	No	No	3.8	166	

	approx_cost(for two people)	listed_in(type)
0	800	Buffet
1	800	Buffet
2	800	Buffet
3	300	Buffet
4	600	Buffet

Summary the dataset

In [5]: `df.info()`

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   name                                  148 non-null    object
1   online_order                         148 non-null    object
2   book_table                           148 non-null    object
3   rate                                 148 non-null    float64
4   votes                                148 non-null    int64
5   approx_cost(for two people)          148 non-null    int64
6   listed_in(type)                      148 non-null    object
dtypes: float64(1), int64(2), object(4)
memory usage: 8.2+ KB
```

conclusion - there is no null value in dataset.

1. What type of restaurant do the majority of customers order from?

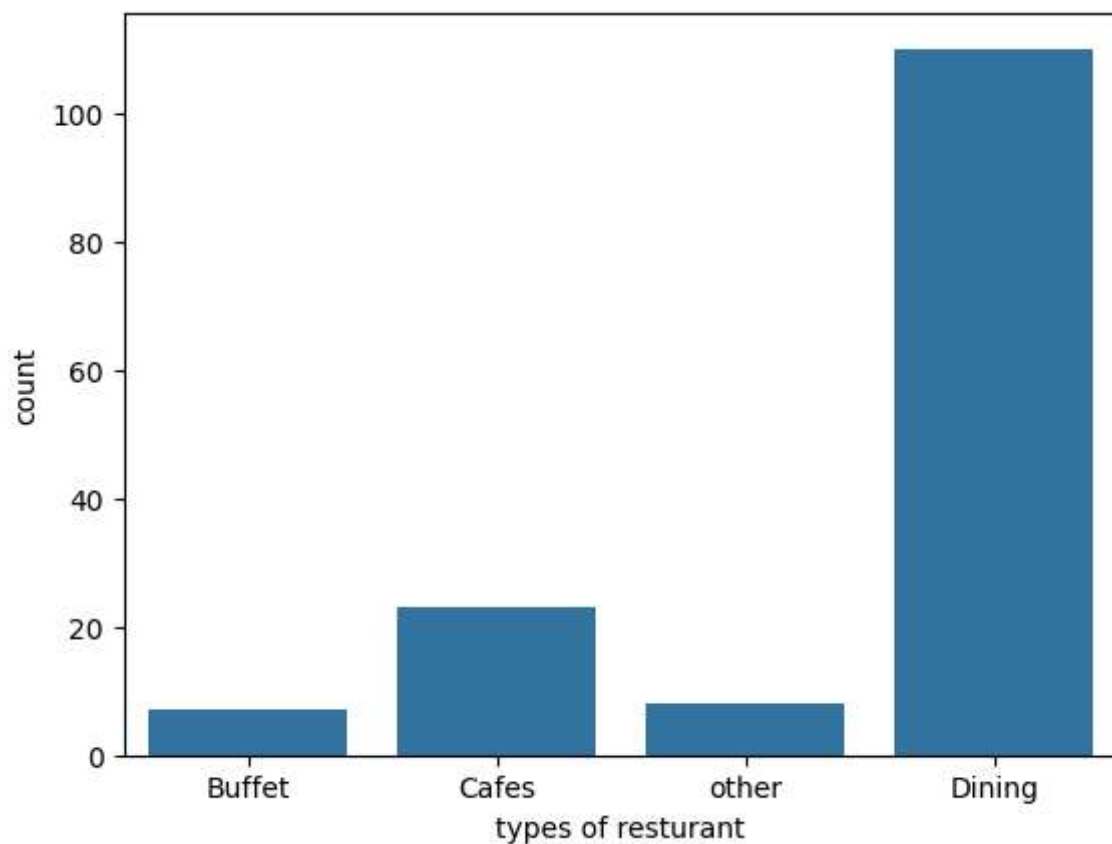
In [6]: `df.head()`

Out[6]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

```
In [7]: sns.countplot(x=df['listed_in(type)'])  
plt.xlabel("types of resturant")
```

Out[7]: Text(0.5, 0, 'types of resturant')



conclusion - Majority of the resturant falls in dining category.

2. How many votes has each type of restaurant received from customers?

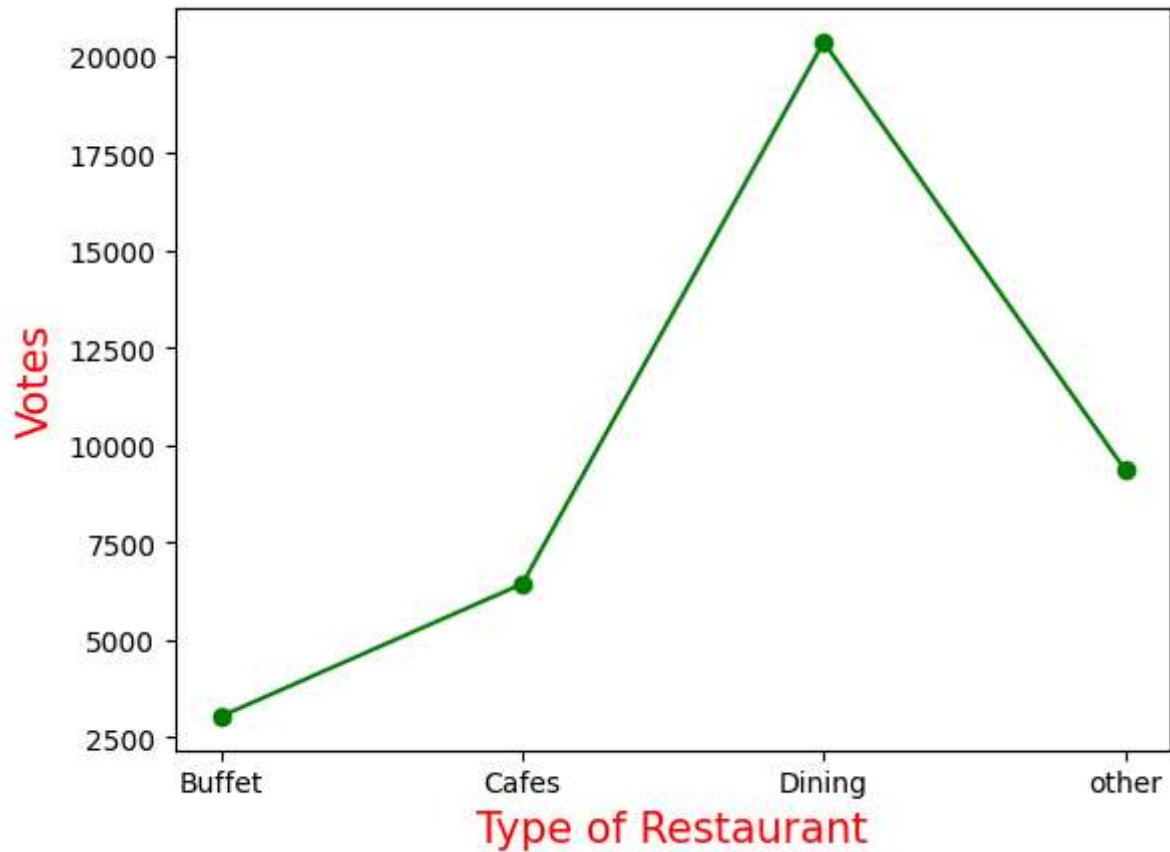
```
In [8]: df.head()
```

```
Out[8]:
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

```
In [9]: grouped_data=df.groupby('listed_in(type)')['votes'].sum()
result=pd.DataFrame({'votes':grouped_data})
plt.plot(result,c="green",marker="o")
plt.xlabel("Type of Restaurant",c="red",size=15)
plt.ylabel("Votes",c="red",size=15)
```

```
Out[9]: Text(0, 0.5, 'Votes')
```



conclusion - Dinning type restaurants has received maximum numbers of votes.

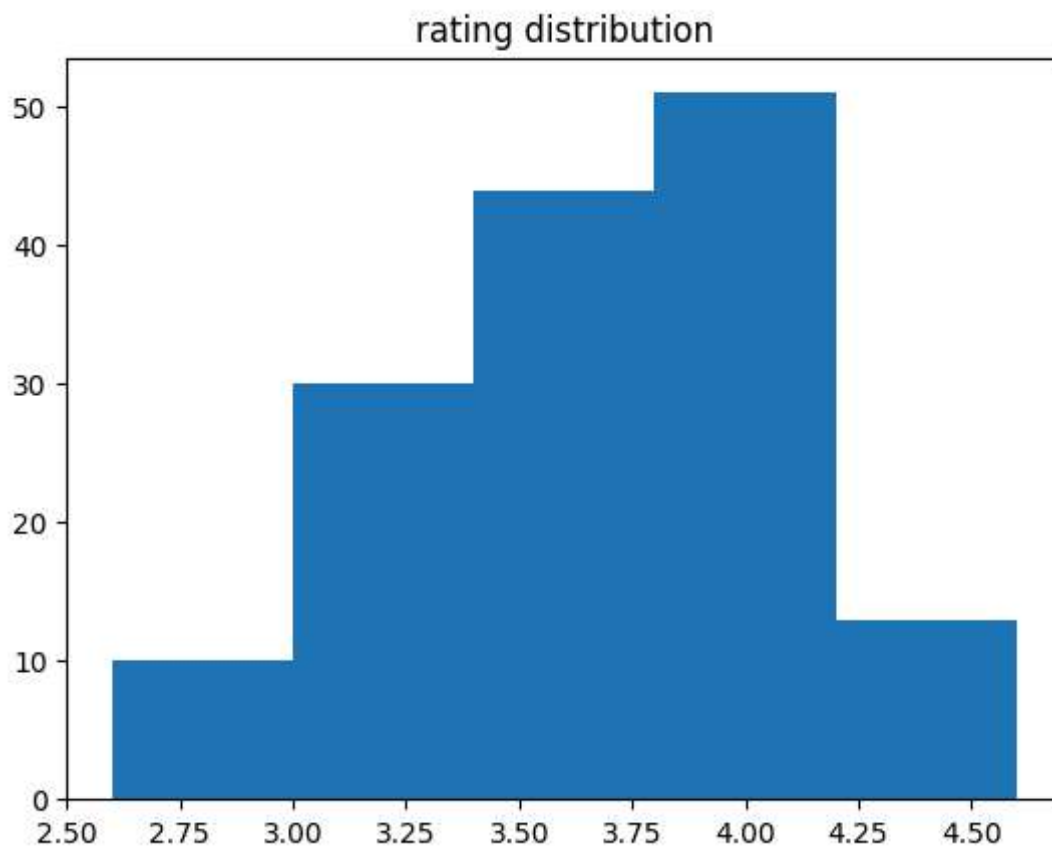
3.What are the rating that the majority of restaurants have received?

```
In [10]: df.head()
```

Out[10]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

```
In [11]: plt.hist(df['rate'],bins=5)
plt.title("rating distribution")
plt.show()
```



conclusion - The majority restaurants received ratings from 3.5 to 4

4. Zomato has observed that most couples order most of their food online. What is their average spending on each order?

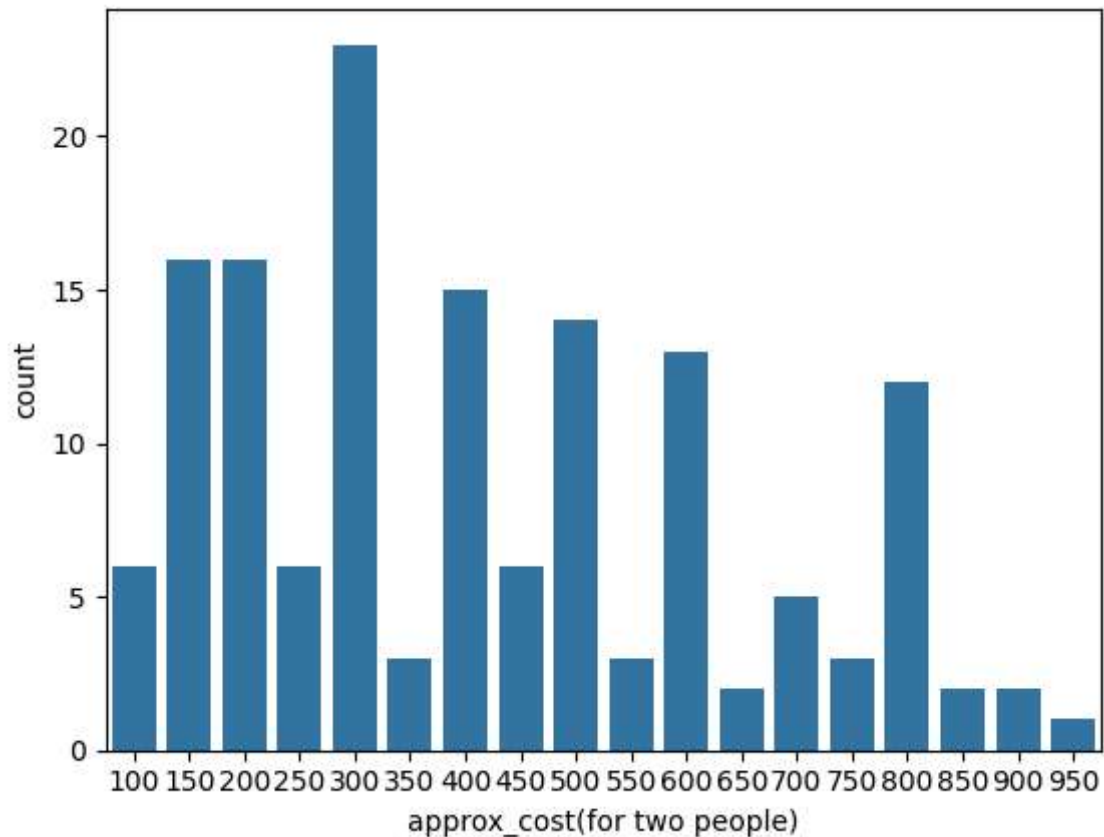
```
In [12]: df.head()
```

```
Out[12]:
```

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

```
In [13]: couple_data=df['approx_cost(for two people)']  
sns.countplot(x=couple_data)
```

```
Out[13]: <Axes: xlabel='approx_cost(for two people)', ylabel='count'>
```



conclusion - The majority of couple prefer restaurants with an approximate cost of 300 rupees.

In [14]: `# 5.which mode (online or offline) has received the maximum rating?`

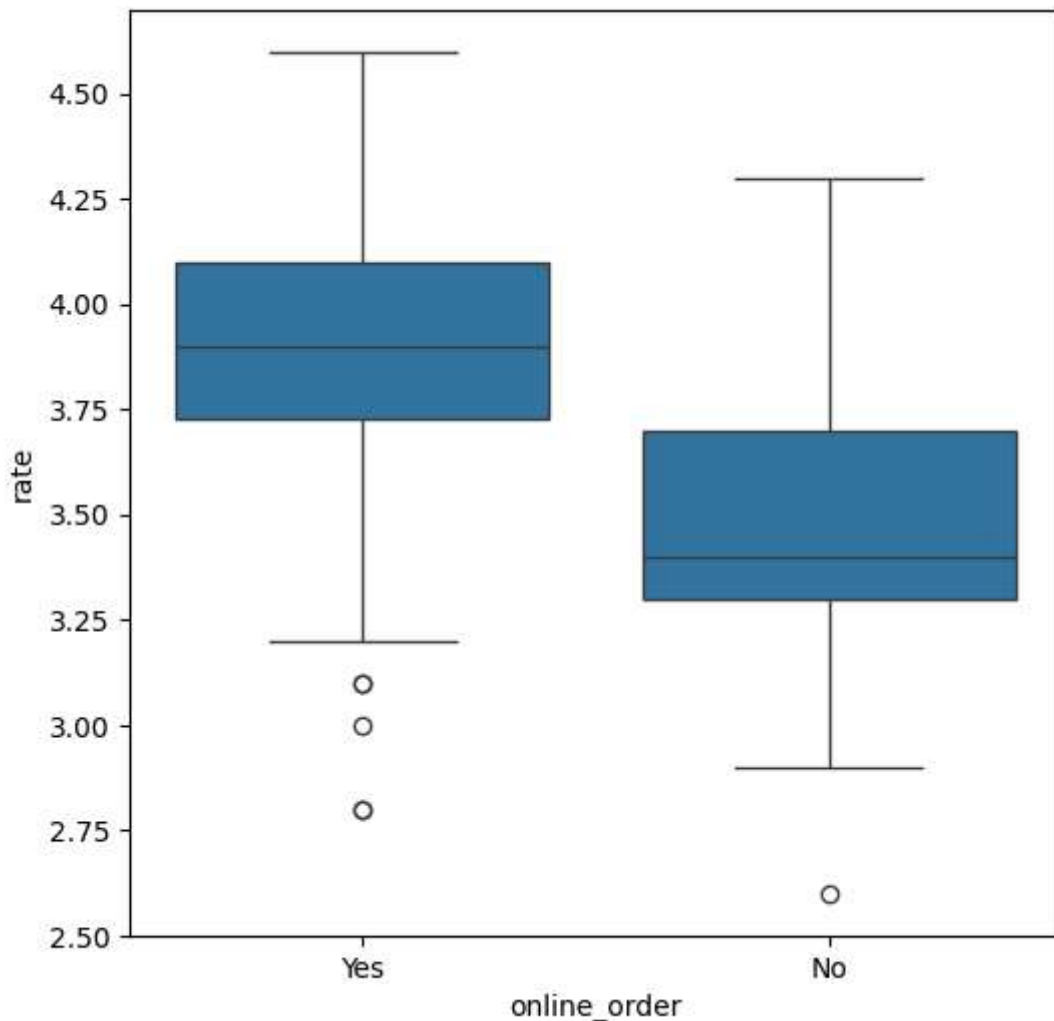
In [15]: `df.head()`

Out[15]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

```
In [16]: plt.figure(figsize=(6,6))  
sns.boxplot(x='online_order',y='rate',data=df)
```

```
Out[16]: <Axes: xlabel='online_order', ylabel='rate'>
```



conclusion - offline order received lower rating in comparison to online order.

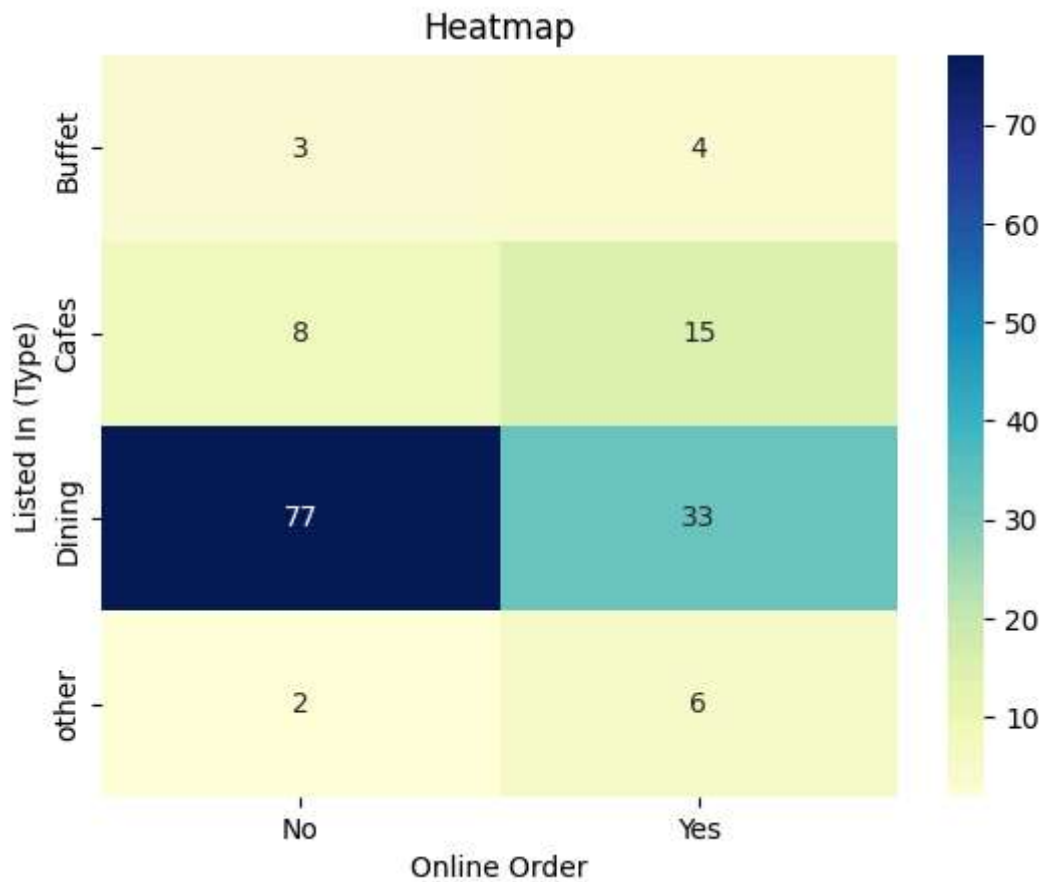
6.Which type of restaurant received more offline orders, so that Zomato can provide those customers with some good offers?

```
In [17]: df.head()
```

Out[17]:

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1	775	800	Buffet
1	Spice Elephant	Yes	No	4.1	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7	88	300	Buffet
4	Grand Village	No	No	3.8	166	600	Buffet

```
In [19]: pivot_table=df.pivot_table(index='listed_in(type)',columns='online_order',aggfunc='
sns.heatmap(pivot_table,annot=True,cmap="YlGnBu",fmt='d')
plt.title("Heatmap")
plt.xlabel("Online Order")
plt.ylabel("Listed In (Type)")
plt.show()
```



conclusion - Dining restaurants primarily accept offline orders, whereas cafes primarily receive online orders. This suggests that clients prefer to place orders in person at restaurants, but prefer online ordering at cafes.

In []: